PART 1 - GENERAL

1.1 Summary

26 13 26 Part 1.1.1.A This section specifies the requirements for factory-assembled, metal-enclosed, medium voltage switchgear for the 11 kV power distribution system of the Meridian Data Centre Campus, Phase 1.

1.2 References

26 13 26 Part 1.2.1 All medium voltage switchgear assemblies shall comply with UL 1558 (Standard for Metal-Enclosed Low-Voltage Power Circuit Breaker Switchgear) and shall be UL listed and labeled. Switchgear components shall be UL recognized or listed as applicable.

26 13 26 Part 1.2.2.A Additional applicable standards include: IS 3427 (Metal-Enclosed Switchgear), IS/IEC 62271-200 (AC Metal-Enclosed Switchgear for Rated Voltages Above 1 kV and Up To and Including 52 kV), IEC 62271-1 (Common Specifications for High-Voltage Switchgear).

1.3 Submittal Requirements

26 13 26 Part 1.3.1.A Submittals shall include general arrangement drawings, single line diagrams, control schematics, busbar current rating calculations, type test reports, and a clause-by-clause compliance matrix.

1.4 Quality Assurance

26 13 26 Part 1.4.1.A Switchgear manufacturer shall have a minimum of 15 years experience in manufacturing medium voltage switchgear assemblies rated 12 kV and above.

26 13 26 Part 1.4.2 Manufacturer shall provide ASTA or KEMA type test certificates for the switchgear assemblies demonstrating compliance with IEC 62271-200. Type test reports shall cover rated voltage, short circuit withstand, temperature rise, internal arc, and dielectric tests.

26 13 26 Part 1.4.3.A Manufacturing facility shall hold current ISO 9001 certification. The Consultant reserves the right to conduct a factory inspection prior to dispatch.

26 13 26 Part 1.4.4.A Routine factory tests shall be conducted on each switchgear panel prior to dispatch, including power frequency voltage withstand, mechanical operation, and wiring checks per IEC 62271-200 Clause 8.

26 13 26 Part 1.4.5 Switchgear shall be rated for continuous operation over an ambient temperature range of -5°C to 45°C. Temperature rise of busbars and connections shall not exceed the limits specified in IEC 62271-200 Table 4 at 45°C ambient.

PART 2 - PRODUCTS

2.1 Acceptable Manufacturers

26 13 26 Part 2.1.1.A Acceptable switchgear manufacturers: Trident Switchgear, or approved equivalent meeting all performance requirements of this section.

2.2 Electrical Ratings

26 13 26 Part 2.2.1.A Switchgear rated voltage shall be 12 kV, suitable for the 11 kV nominal system voltage. Rated insulation level shall be 12/28/75 kV (rated voltage / power frequency withstand / lightning impulse withstand).

26 13 26 Part 2.2.1.B Main busbar continuous current rating shall be a minimum of 2000 A at 45°C ambient temperature.

26 13 26 Part 2.2.2.A Rated short-time withstand current shall be a minimum of 31.5 kA (rms) for a duration of 3 seconds.

26 13 26 Part 2.2.2.B Short-circuit current withstand duration shall be a minimum of 3 seconds. All busbars, connections, and supporting insulators shall be rated accordingly.

26 13 26 Part 2.2.3 Rated short-circuit making capacity shall be a minimum of 66 kA peak. Circuit breakers shall have a making capacity not less than the peak value of the rated short-circuit withstand current.

26 13 26 Part 2.2.4 Rated lightning impulse withstand voltage (BIL) shall be a minimum of 75 kV (1.2/50 microsecond wave). Power frequency withstand voltage shall be 28 kV for 1 minute.

2.3 Construction and Components

26 13 26 Part 2.3.1.A Switchgear enclosure external protection shall be a minimum of IP4X per IEC 60529. Internal compartment protection between functional units shall be IP2X minimum.

26 13 26 Part 2.3.1.B Switchgear construction shall be compartmentalised type with metallic shutters on all circuit breaker compartments, busbar compartments, and cable compartments.

26 13 26 Part 2.3.1.C Circuit breakers shall be vacuum type, withdrawable on truck mechanism. Breaker truck shall have test, connected, and disconnected positions with mechanical interlocks.

26 13 26 Part 2.3.1.D Anti-condensation space heaters shall be provided in each switchgear compartment. Heaters shall be thermostatically controlled and powered from the auxiliary supply.

26 13 26 Part 2.3.1.E Switchgear shall be classified for Internal Arc Containment (IAC) AFLR (Accessibility: Front, Lateral, Rear) rated for 31.5 kA for 1 second per IEC 62271-200 Annex A.

26 13 26 Part 2.3.1.F Loss of service continuity classification shall be LSC-2B per IEC 62271-200, ensuring that maintenance on any single functional unit does not require de-energisation of adjacent units.

26 13 26 Part 2.3.2.A Main busbars shall be high conductivity electrolytic copper, silver-plated at all joint surfaces. Busbars shall be sized for the rated continuous current with temperature rise within IEC 62271-200 limits.

26 13 26 Part 2.3.3.A Cable termination compartment shall accommodate XLPE-insulated cables with heat-shrink terminations. Cable compartment depth shall be a minimum of 600 mm.

26 13 26 Part 2.3.3.B Earthing busbar shall be high conductivity copper, continuous throughout the switchgear lineup, and sized for the full rated short circuit current. Earth bus connections shall be bolted with calibrated torque.

26 13 26 Part 2.3.4.A Circuit breaker operating mechanism shall be spring-charged, motor-wound type suitable for auto-reclosing. Stored energy shall be sufficient for one close-open-close operation.

26 13 26 Part 2.3.5.A Protection relays shall be microprocessor-based numerical type with overcurrent, earth fault, and under/over voltage functions. Relay communications shall be IEC 61850 compatible.

26 13 26 Part 2.3.6.A Metering instruments shall include digital multifunction meters with accuracy class 0.5. Meters shall display voltage, current, power, energy, power factor, and frequency.

26 13 26 Part 2.3.7.A Current transformers shall have metering accuracy class 0.5s and protection accuracy class 5P20. CT ratios shall be as shown on the single line diagram.

26 13 26 Part 2.3.7.B Current transformers shall have a rated burden not less than the connected metering and protection burden with a 25% margin.

26 13 26 Part 2.3.7.C Voltage transformers shall have accuracy class 0.5 for metering and 3P for protection. Primary voltage shall be 11 kV/√3, secondary 110 V/√3.

PART 3 - EXECUTION

3.1 Installation

26 13 26 Part 3.1.1.A Switchgear shall be installed on prepared foundations with adequate clearances per manufacturer recommendations. Minimum front clearance shall be 2000 mm. Minimum rear clearance shall be 1000 mm.

3.2 Field Quality Control

26 13 26 Part 3.2.1.A Perform insulation resistance tests on all busbars and cable terminations prior to energisation. Minimum acceptable IR value is 1000 MΩ at 5 kV DC.